

# Paul O. Hayne

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University of Colorado, Boulder  
Astrophysical & Planetary Sciences Department  
391 UCB  
Boulder, CO 80309

[Paul.Hayne@Colorado.edu](mailto:Paul.Hayne@Colorado.edu)  
<https://www.colorado.edu/aps/paul-hayne>  
Office: Duane Physics D321 || LASP-SPSC N294  
Phone: (303) 735-6399 [O] (720) 390-9276 [M]

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**Summary**      My research focuses on the surfaces and atmospheres of icy planets and moons.  
I develop computational models, missions, and instrumentation to better understand these planetary bodies. Teaching, mentoring, and public engagement are also fundamental to my work.

**Positions Held**      **University of Colorado, Boulder**  
2024 – present *Associate Professor* (with tenure)  
2018 – 2024 *Assistant Professor*  
Astrophysical & Planetary Sciences Department,  
and Laboratory for Atmospheric & Space Physics

**NASA – Jet Propulsion Laboratory**, California Institute of Technology  
2012 – 2017 *Research Scientist*

**California Institute of Technology**  
2011 – 2012 *Postdoctoral Scholar* (Supervisor: O. Aharonson)

**Education**      **University of California, Los Angeles**  
*Ph.D.*, Geophysics & Space Physics, 2010 (Advisor: D. A. Paige)  
  
**Stanford University**  
*M.S.*, Geophysics, 2005 (Advisors: N. H. Sleep & J. J. Lissauer)  
*B.S.*, Geophysics, 2003

**Mission Experience**      **Emirates Mission to the Asteroid belt (EMA)**, UAE Space Agency  
and Laboratory for Atmospheric & Space Physics  
*Project Scientist* (2021–present)

**NASA Lunar Compact Infrared Imaging System (L-CIRiS)**  
*Principal Investigator* (2019–present), CLPS-CP22 lunar south polar mission

**NASA Artemis Science Team**  
*Co-Investigator* (2026–present), First Crewed South Polar Mission

**NASA Lunar-Vulkan Imaging and Spectroscopy Explorer (Lunar-VISE)**  
*Co-Investigator* (2019–present)  
*Investigation Lead*, Lunar-VISE Compact Infrared Imaging System (LV-CIRiS)

**NASA Europa Clipper**  
*Co-Investigator* (2017–present) and *Investigation Scientist* (2015–2018)  
Europa Thermal Emission Imaging System (E-THEMIS)

**NASA Janus**  
*Co-Investigator* and *Investigation Lead* (2019–2024)  
Thermal IR Camera

**NASA Mars Reconnaissance Orbiter**  
*Co-Investigator* (2017–present)  
Mars Climate Sounder

|                                        |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mission<br>Experience<br>(cont.)       | <b>NASA Lunar Flashlight</b><br><i>Co-Investigator</i> (2012–2023)                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                  |
|                                        | <b>NASA Lunar Reconnaissance Orbiter</b><br><i>Co-Investigator</i> (2011–present)<br>Diviner Lunar Radiometer Experiment<br><br><i>Science Team Affiliate:</i> <b>NASA CLPS/IM-2 South Pole</b> Moon3D camera (2024–present)<br><b>NASA OSIRIS-REx</b> (2018–2020), <b>NASA Dawn</b> (2016–2019),<br><b>NASA/ESA Cassini-Huygens</b> (2006–2014) |                                                                                                                                                                                                                                                  |
| Selected<br>External<br>&<br>Contracts | <b>Total funding received: ~\$14.0M (PI: ~\$5.7M; Co-I/subaward: ~\$8.2M)</b>                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                  |
|                                        | 2026–2029                                                                                                                                                                                                                                                                                                                                        | <i>Supporting Artemis III Science Operations through Thermal Modeling and Volatiles Expertise</i><br>PI: P. Hayne<br>NASA Artemis III Science Team / \$284k                                                                                      |
|                                        | 2025–2027                                                                                                                                                                                                                                                                                                                                        | <i>Advanced 3-d Thermophysical Model to Enable Europa Clipper Interdisciplinary Science</i><br>PI: P. Hayne<br>NASA Precursor Science Investigations for Europa<br>\$261k                                                                        |
|                                        | 2025–2027                                                                                                                                                                                                                                                                                                                                        | <i>Topographic Modeling to Simulate Multi-scale Roughness for Europa Clipper Investigations and Science</i><br>Co-I: P. Hayne<br>NASA Working Models for Europa Clipper / JPL / \$75k                                                            |
|                                        | 2025–2026                                                                                                                                                                                                                                                                                                                                        | <i>Moon3D Thermophysical Modeling</i><br>Co-I: P. Hayne<br>NASA RESOURCE: Resource Exploration on the Moon / ARC / \$90k                                                                                                                         |
|                                        | 2025–2026                                                                                                                                                                                                                                                                                                                                        | <i>LunarAssay via Small Satellite Orbiter Phase 1A</i><br>PI: P. Hayne<br>DARPA LASSO Payload / \$299k                                                                                                                                           |
|                                        | 2019–2028                                                                                                                                                                                                                                                                                                                                        | <i>Lunar Vulkan Imaging and Spectroscopy Explorer (Lunar-VISE)</i><br>PI: K. Donaldson Hanna (Co-I/Instrument Lead: P. Hayne)<br>NASA Payloads and Research Investigations on the Surface of the Moon (PRISM)<br>\$35M (\$2.2M subaward to LASP) |
|                                        | 2019–2028                                                                                                                                                                                                                                                                                                                                        | <i>Lunar Compact Infrared Imaging System (L-CIRiS)</i><br>PI: P. Hayne<br>NASA Lunar Science and Instrument Technology Payloads / \$4.1M                                                                                                         |
|                                        | 2015–2034                                                                                                                                                                                                                                                                                                                                        | <i>Europa Thermal Emission Imaging System</i><br>PI: P. Christensen (Co-I: P. Hayne)<br>NASA Europa Clipper Mission / ~\$600k subaward to LASP for Phases D–E                                                                                    |
|                                        | 2016–2025                                                                                                                                                                                                                                                                                                                                        | <i>Lunar Reconnaissance Orbiter Extended Missions 3–5</i><br>PI: D. Paige (Co-I: P. Hayne)<br>NASA Discovery Program / ~\$2.2M allocation for Co-I Hayne                                                                                         |

|                   |           |                                                                                                                                                                                                                              |
|-------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grants<br>(cont.) | 2016–2025 | <i>Mars Reconnaissance Orbiter Extended Missions 4–6</i><br>PI: A. Kleinböhl (Co-I: P. Hayne)<br>NASA Mars Program / ~\$1.5M allocation for Co-I Hayne                                                                       |
|                   | 2019–2024 | <i>Janus Mission</i><br>PI: D. Scheeres (Co-I/Instrument Lead: P. Hayne)<br>NASA SIMPLEX Program / ~ \$400k subaward to LASP                                                                                                 |
|                   | 2013–2023 | <i>Lunar Flashlight Science Investigation</i><br>PI: B. Cohen (Co-I: P. Hayne)<br>NASA Advanced Exploration Systems / ~\$900k allocation for Co-I Hayne                                                                      |
|                   | 2018–2021 | <i>Boulders on Bennu: Modeling Thermal Emission from Boulders for Yarkovsky Effect and Thermal Inertia Investigations</i><br>PI: P. Hayne (Science PI: C. Elder)<br>NASA OSIRIS-REx Participating Scientist Program / \$346k |
|                   | 2017–2018 | <i>Unlocking the Climate Record Stored within Mars’ Polar Layered Deposits</i><br>PI: P. Hayne<br>W. M. Keck Institute for Space Studies / \$50k                                                                             |
|                   | 2015–2018 | <i>Inter-seasonal and Inter-annual Surface Dust Fluxes on Mars</i><br>PI: P. Hayne (Science PI: S. Piqueux)<br>NASA Mars Data Analysis Program / \$331k                                                                      |
|                   | 2013–2017 | <i>Volatile Regolith Thermal Investigation Consortium for Exploration and Science (VORTICES)</i><br>PI: A. Rivkin (Co-I: P. Hayne)<br>NASA Solar System Exploration Research Virtual Institute / ~\$120k                     |
|                   | 2014–2016 | <i>Thermal Infrared Observations of the Moon during Lunar Eclipse</i><br>PI: P. Lucey (Co-I: P. Hayne)<br>NASA Planetary Astronomy / \$143k allocation for P. Hayne                                                          |
|                   | 2013–2014 | <i>New Approaches to Lunar Ice Detection and Mapping</i><br>PI: P. Hayne<br>W. M. Keck Institute for Space Studies / \$50k                                                                                                   |

National Service    **Europa ICONS**  
*Member of NASA's Program Definition Tiger Team (2023–2024)*

**NASA Artemis**  
*Course Developer, Astronaut Training – Lunar Surface Environment (2024–present)*

**International Workshop on Instrumentation for Planetary Missions**  
*Science Organizing Committee (2025)*

**Keck Institute for Space Studies**, California Institute of Technology  
*Study Co-Lead*  
     2017–2018: [Unlocking the Climate Record Stored within Mars' Polar Layered Deposits](#)  
     2013–2014: [New Approaches to Lunar Ice Detection and Mapping](#)

**American Astronomical Society**, Division for Planetary Sciences  
*Editorial Board Member, Icarus (2020–present)*  
*Member of the Federal Relations Subcommittee (2017–2022)*  
*Local Organizing Committee, Annual Meeting (2016)*  
*Panelist, Annual Meeting (2015)*

**American Geophysical Union**  
*Session Co-organizer (Mars Atmosphere), Annual Meeting (2014–2024)*

**Early-Career Planetary Explorers**  
*Co-founder (2012) and Co-leader (2012–2020)*  
     World-wide advocacy organization for early-career planetary scientists

**NASA Science Mission Directorate**  
*Appointed Member, Mars Exploration Program:*  
     Ice and Climate Evolution Science Assessment Group (2018–2019)  
*Review Panel Member (2012–present)*

**National Research Council – Space Studies Board**  
*Raconteur, Planetary Science Decadal Survey (2009–2010)*

**Referee** for major scientific journals:  
*Journal of Geophysical Research, Geophysical Research Letters, Icarus, Astrophysical Journal, Planetary & Space Science, Advances in Space Research, Space Science Reviews, Geology, Science Advances, Nature Astronomy, Nature, PNAS*

Departmental  
&  
LASP  
Service

*Committee Membership (APS Department):*

|           |                                                       |
|-----------|-------------------------------------------------------|
| 2025/2026 | Tenure-track Faculty Search Committee                 |
| 2025/2026 | Graduate Curriculum and Concerns Committee            |
| 2025      | Primary Unit Evaluation Committee (Kazachenko)        |
| 2024/2025 | McCray Prize Postdoctoral Fellowship Committee        |
| 2024/2025 | Sommers-Bausch Observatory Advisory Committee (Chair) |
| 2024/2025 | Faculty Search Committee                              |
| 2024/2025 | Graduate Curriculum and Concerns Committee            |
| 2024/2025 | Beyond Boulder (Graduate School Info Session)         |
| 2023/2024 | Graduate Admissions Committee (Vice Chair)            |
| 2023/2024 | Faculty Search Committee                              |
| 2023/2024 | Social Committee                                      |
| 2022/2023 | Outreach & Development Committee                      |
| 2022/2023 | Friday Lunch Seminar Committee                        |
| 2022/2023 | Graduate Admissions Committee (Chair)                 |
| 2021/2022 | Graduate Admissions Committee                         |
| 2021/2022 | Faculty Search Committee                              |
| 2021/2022 | Outreach & Development Committee                      |
| 2020/2021 | Fall Semester/COVID Planning Committee                |
| 2020/2021 | Grad Welcome and Social Committee                     |
| 2019/2020 | Social Committee (Chair)                              |
| 2019/2020 | Colloquium Committee                                  |
| 2019/2020 | Undergraduate Curriculum and Concerns Comm.           |
| 2019/2020 | Graduate Admissions Committee                         |
| 2018/2019 | Social Committee (Acting Chair)                       |
| 2018/2019 | Colloquium Committee                                  |

*Committee Membership (LASP):*

|              |                                                  |
|--------------|--------------------------------------------------|
| 2023–present | LASP Research Scientist promotion committees     |
| 2025         | LASP Strategic Planning Committee                |
| 2024–2025    | LASP Internal Research and Development Committee |
| 2023–2024    | LASP Strategic Planning Team                     |

*Departmental Mentoring:*

|              |                                             |
|--------------|---------------------------------------------|
| 2018–present | APS Department Undergraduate Student Mentor |
| 2018–present | APS Department Graduate Student Mentor      |

*Additional Service:*

|           |                                                |            |
|-----------|------------------------------------------------|------------|
| 2018-2019 | Co-organizer: “Planet Lunch”                   | CU Boulder |
| 2018-2020 | Organizer: Natural Sciences New Faculty Coffee | CU Boulder |

Visiting  
Appointments

**Lawrence Livermore National Laboratory**

*Visiting Scientist* (2017)

Planetary Defense Program / Weapons and Complex Integration (WCI)

**Weizmann Institute of Science, Israel**

*Visiting Scientist* (2015, 2016)

The Helen Kimmel Center for Planetary Science

|                    |      |                                                                                |
|--------------------|------|--------------------------------------------------------------------------------|
| Honors<br>& Awards | 2022 | <i>RIO Faculty Fellow</i> , CU Research & Innovation Office                    |
|                    | 2013 | <i>NASA Group Achievement Award</i> , Diviner Lunar Radiometer science team    |
|                    | 2011 | <i>NASA Group Achievement Award</i> , Mars Climate Sounder science team        |
|                    | 2010 | <i>Best Student Paper Award</i> , NASA Lunar Science Forum                     |
|                    | 2010 | <i>NASA Group Achievement Award</i> , Diviner Lunar Radiometer operations team |
|                    | 2010 | <i>NASA Group Achievement Award</i> , Diviner Lunar Radiometer science team    |
|                    | 2008 | <i>Simon Latimer Award for Service</i> , UCLA                                  |
|                    | 2006 | <i>Graduate Fellowship</i> , Institute of Geophysics and Planetary Physics     |
|                    | 2006 | <i>Chancellor's Prize</i> , UCLA                                               |
|                    | 2003 | <i>Graduate Fellowship</i> , Dept. of Geophysics, Stanford University          |
|                    | 2001 | <i>Summer Research Fellowship</i> , Dept. of Physics, Stanford University      |

|                                     |                                             |                    |                             |
|-------------------------------------|---------------------------------------------|--------------------|-----------------------------|
| Exam<br>Committees<br>&<br>Defenses | <i>Graduate Comprehensive Examinations:</i> |                    |                             |
|                                     | 2025                                        | Mirielle Caradonna | CU / APS                    |
|                                     | 2024                                        | James Crowley      | CU / APS                    |
|                                     | 2024                                        | Michael Rothman    | CU / APS                    |
|                                     | 2024                                        | Sam Cartwright     | CU / GEOL                   |
|                                     | 2023                                        | Robert Stevens*    | CU / APS *Primary Advisor   |
|                                     | 2023                                        | Helle Skjetne      | CU / GEOL                   |
|                                     | 2023                                        | Kris Laferriere    | Purdue U.                   |
|                                     | 2022                                        | Harry Brodsky      | CU / GEOL                   |
|                                     | 2022                                        | Patrick Behr       | CU / APS                    |
|                                     | 2022                                        | Dahlia Baker       | CU / AES                    |
|                                     | 2021                                        | Kya Sorli*         | CU / APS *Primary Advisor   |
|                                     | 2020                                        | E. Fernando Cruz   | CU / APS Non-advocate Chair |
|                                     | 2020                                        | Rachel Hoover      | CU / GEOL                   |
|                                     | 2019                                        | Eryn Cangi         | CU / APS                    |
|                                     | 2019                                        | Noora Alsaeed      | CU / APS Non-advocate Chair |
|                                     | 2019                                        | Raven Larson*      | CU / APS *Primary Advisor   |
|                                     | 2019                                        | Andrew Wilcoski*   | CU / APS *Primary Advisor   |
|                                     | 2019                                        | Perianne Johnson   | CU / APS Non-advocate Chair |
|                                     | 2019                                        | William Waalkes    | CU / APS                    |
|                                     | 2018                                        | Momchil Molnar     | CU / APS                    |
|                                     | 2018                                        | Arika Egan         | CU / APS                    |

*Ph.D. Dissertation Defenses:*

|      |                             |                           |                  |
|------|-----------------------------|---------------------------|------------------|
| 2025 | <b>Dr. Kya Sorli*</b>       | CU / APS                  | *Primary Advisor |
| 2025 | <b>Dr. Heshani Pieris*</b>  | CU / APS                  | *Primary Advisor |
| 2024 | Dr. Koundinya (Ken) Kuppa   | CU / AES                  |                  |
| 2024 | Dr. Lucas Lange             | Sorbonne Université / LMD |                  |
| 2022 | <b>Dr. Noora Alsaeed*</b>   | CU / APS                  | *Primary Advisor |
| 2022 | <b>Dr. Andrew Wilcoski*</b> | CU / APS                  | *Primary Advisor |
| 2022 | Dr. Chase Chivers           | Georgia Tech / EAS        |                  |
| 2022 | Dr. Christopher Kremer      | Brown U. / DEEPS          |                  |
| 2019 | Dr. Marek Slipski           | CU / GEOP                 |                  |

*Undergraduate Honors Thesis Defenses:*

|      |                          |           |                  |
|------|--------------------------|-----------|------------------|
| 2019 | Carlos Eytan Gary Bicas* | CU / APS  | *Primary Advisor |
| 2018 | Amanda Alexander         | CU / GEOL |                  |

Teaching **University of Colorado Boulder**

*ASTR 1030: Accel. Introductory Astronomy* (Fall, 2018)

Instructor for undergraduate majors course (115 students)

*ASTR 3400: Research Methods* (Spring, 2023; Spring, 2024)

Instructor for undergraduate majors course (30 students)

*ASTR 3720: Planets and their Atmospheres* (Spring, 2018)

Instructor for upper-division undergraduate course (84 students)

*ASTR 3750: Planets, Moons, and Rings* (Fall, 2020; Fall, 2021; Fall, 2022; Fall, 2024)

Instructor for undergraduate majors course (~100 students)

*ASTR 5800: Planetary Surfaces & Interiors* (Fall, 2019; Fall, 2021; Fall, 2023)

Instructor for graduate course (~15 students)

*ASTR 5810: Planetary Atmospheres* (Spring, 2026)

Instructor for graduate course (~15 students)

*ASTR 5830: Remote Sensing of Planetary Surfaces and Atmospheres* (Spring, 2025)

Instructor for graduate course (~10 students)

*ASTR 5835: Seminar in Planetary Science* (Spring, 2023; Fall, 2024)

Co-Instr. for grad. seminar (~25 students): “Asteroids” and “Ocean Worlds”

**California Institute of Technology**

*Ge 151: Planetary Surfaces* (2011)

Co-Instructor & Lecturer for graduate course (~6 students)

**University of California, Los Angeles**

*ESS 10: Exploring Mars* (2008)

Teaching Assistant and Lecturer for undergraduate course (60 students)

**Stanford University**

*Geophysics 150: Physics of the Earth* (2004, 2005)

Teaching Assistant for graduate course (~6 students)

*Physics 50: Observational Astronomy* (2002, 2003)

Teaching Assistant for undergraduate course (~20 students)

## Mentoring **Laboratory for Atmospheric and Space Physics**

### *Postdocs*

2023–present: Dr. Harish  
2022–present: Dr. Patrick O’Brien  
2019–2020: Dr. Margaret Landis (now: faculty member, ASU)

### *Graduate Students*

2024–present: Rhali Attar (APS, CU)  
2022–present: Robert Stevens (APS, CU)  
2023–2025: **Heshani Pieris, *Ph.D.*** — ‘**25** (APS, CU → LASP Researcher)  
2020–2025: **Kya Sorli, *Ph.D.*** — ‘**25** (APS, CU → U. Oxford) \*NSF Fellow  
2020–2022: **Noora Alsaeed, *Ph.D.*** — *Dec.* ‘**22** (APS, CU)  
2018–2022: **Andrew Wilcoski, *Ph.D.*** — *Dec.* ‘**22** (APS, CU)  
2019–2020: **Raven Larson, *M.S.*** — *May* ‘**20** (APS, CU)

### *Undergraduate Students*

2024–present: Maya Nussbaumer (APS Dept., CU Boulder)  
2024–2025: Bethany Fuhrman (CU Boulder, B.S. → U. Hawaii, Ph.D.)  
2024 (summer): Matthew Perkins (NASA Europa ICONS)  
2023 (summer): Fahad Alawadhi (AES Dept., CU Boulder)  
2023 (summer): Alia Almansoori (U. Edinburgh)  
2023 (summer): Tyler Jx Bellison (Fort Lewis College)  
2022–2024: Isabella Wise (APS Dept., CU Boulder)  
2022–2023: Sam Aberbook (Physics Dept., CU Boulder)  
2018–2021: Tara Tomlinson (CU Boulder, B.S. → Dartmouth College, Ph.D.)  
2019–2020: Shaelyn Raposa (CU Boulder, B.S. → N. Ariz. U., Ph.D.)  
2018–2019: Carlos Eytan Gary Bicas (CU Boulder, B.S. → Stony Brook U., Ph.D.)  
2018–2019: Tyler Horvath (CU Boulder, B.S. → UCLA, Ph.D.)

### **Caltech Summer Undergraduate Research Fellows (SURF)**

2017: James Haber (Cornell University, B.S. → Purdue U., Ph.D.)  
2015: Léa Bonnefoy (Cornell University, B.S. → U. Arizona, Ph.D.)  
2014: Clifford Watkins (Carleton College, B.S. → U. Hawaii, Ph.D.)  
2011: Michael Lauria (Caltech, B.S.)

### **Maximizing Student Potential (MSP)**

2015–2018: José Martínez Camacho (Citrus College → SMU, Ph.D.)

### **Caltech Postdoctoral Fellows at JPL**

2017–2018: Dr. Quentin Vinckier (now: JPL Technologist)  
2016–2018: Dr. Catherine M. Elder (now: JPL Scientist)

## Invited Talks

2026 NASA Artemis Science Series  
 2025 NASA Europa Clipper Science Series  
 2025 National Academy of Sciences / U.S.-Africa Frontiers Symposium  
 2025 L'Space / Arizona State University  
 2024 L'Space / Arizona State University  
 2023 Brown University (postponed)  
 2021 Purdue University  
 2020 Planetary Science Decadal Survey  
 2020 Observatoire de la Côte d'Azur  
 2020 NASA Lunar Surface Science Workshop  
 2019 University of California, Los Angeles  
 2018 Applied Physics Laboratory, Johns Hopkins University  
 2017 Lawrence Livermore National Laboratory  
 2017 University of Colorado, Boulder  
 2016 Lunar & Planetary Institute  
 2016 Weizmann Institute of Science (Israel)  
 2015 International Space Exploration Coordination Group,  
 2015 NASA Exploration Science Forum  
 2015 University of California - Santa Cruz  
 2014 SETI Institute  
 2013 Lunar & Planetary Institute  
 2012 California Institute of Technology  
 2012 NASA – Jet Propulsion Laboratory

## Outreach & Public Engagement

### CU Outreach Grant

*Faculty Award Co-Recipient*, “Engaging Underserved Students with  
 CU’s Cutting-edge Space Research” (2021–2022)

### Ad Astra Academy

*Co-founder and Co-leader*, 2015–present  
 Brazil (2015, 2017), Kenya (2016), Colorado (2021, 2024)  
 Science and exploration program for underserved student populations

## Media Interviews & Popular Writing

*Space.com*: [Sunny days on Mars come with a weather warning](#) (Dec. 2024)  
 Also reported in: *Forbes* and other media outlets  
*The Conversation*: [Scientists suspect there’s ice hiding on the Moon](#) (Nov. 2023)  
 Reprinted in: *5 Essential Reads on Modern Lunar Missions* (Dec. 2023)  
*Popular Science*: [We finally have a detailed map of water on the moon](#) (Mar. 2023)  
*The Atlantic*: [It snows on Mars](#) (Jan. 2023)  
*Science Daily*: [... Sweater weather on the Moon](#) (Jul. 2022)  
*The Conversation*: [A rocket crashes into the Moon](#) (Mar. 2022)  
*Space.com*: [How a rocket crash on the Moon will help impact physics](#) (Mar. 2022)  
*CNN*: [Ancient volcanoes ... a rare resource for lunar explorers](#) (May 2022)  
*Wired*: [A clever way to map the Moon’s surface: using shadows](#) (Oct. 2022)  
*The New York Times*: [Life on the planet Mercury?](#) (Mar. 2020)  
*Smithsonian*: [The Moon has more hidden water](#) (Oct 2020)  
*Popular Science*: [Europa is the new Mars](#) (Jun. 2013)

*Quirks & Quarks* (podcast interview):

[The Moon’s a wetter place than we thought](#) (Oct. 2020)

*AirTalk* (radio interview):

[Studies confirm longstanding theory that there’s water on the Moon](#) (Oct. 2020)

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| Professional<br>Affiliations | American Astronomical Society - Division for Planetary Sciences<br>American Geophysical Union<br>European Geosciences Union<br>Asia Oceania Geosciences Society                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Skills<br>& Training         | <p><b>Remote sensing &amp; spacecraft instrumentation,<br/>Modeling and numerical simulation</b></p> <p><b>Computer programming languages</b><br/> Proficient/expert: C/C++, Python, MATLAB, IDL<br/> Familiar: Fortran 90/95, shell scripting, etc.</p> <p><b>Mission formulation experience</b><br/> <i>Team-X</i>, NASA Jet Propulsion Laboratory<br/> <i>Team-A</i>, NASA Jet Propulsion Laboratory<br/> <i>Capture Lead Training</i>, NASA Jet Propulsion Laboratory<br/> Co-I and Deputy PI on mission proposals to NASA's <i>Discovery</i> and <i>New Frontiers</i><br/> Successful PI on multiple instrument proposals to NASA mission programs</p> <p><b>Additional training</b><br/> <i>NASA Astrobiology Winter School and Field Course</i> (2011)<br/> <i>International Astrobiology Summer School</i> (2008)<br/> <i>PADI Open Water Diver</i> scuba certification (50+ dives)<br/> <i>Eagle Scout</i> (1998), BSA Troop 676, Issaquah, WA</p> |

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## Paul O. Hayne: Refereed Journal Articles

**Number of peer-reviewed publications: 111** (as of April, 2026)

First-author: 10 / Student-supervised first-author: 16 / Postdoc-supervised first-author: 11

*h-index*: 36 ([ADS](#)), 40 ([Google Scholar](#))

*i100-index* (# papers cited  $\geq 100$  times): 14 ([ADS](#)), 20 ([Google Scholar](#))

Total citations: 5,107 ([ADS](#)), 7,619 ([Google Scholar](#))

(\*) indicates ([U,G](#)) **student** or ([P](#)) **postdoc** supervised or co-supervised by Hayne

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1. Aharonson, O., **Hayne, P. O.**, & Schörghofer, N. (2026), Observational constraints on the history of lunar polar ice accumulation, *Nature Astronomy*, doi: [10.1038/s41550-026-02822-9](https://doi.org/10.1038/s41550-026-02822-9).
2. Williams, J-P., Valencia, S., Bennett, K. A., et al. (incl. **Hayne, P. O.**) (2026), Morphometric Properties of the CP-21 Landing Site on the Moon at Mons Gruithuisen Gamma, *Planet. Sci. J.*, 7(4), 78, doi: [10.3847/PSJ/ae523b](https://doi.org/10.3847/PSJ/ae523b).
3. Amaro, B., Prieur, N., Rubanenko, L., et al. (incl. **Hayne, P. O.**) (2026), Effect of Boulder-Size Distributions on Thermally Derived Rock Abundances on the Moon, *J. Geophys. Res.*, 131(3), e2024JE008769, doi: [10.1029/2024JE008769](https://doi.org/10.1029/2024JE008769).
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